

HARNETT REGIONAL, NC, USA

WMO: 746936

Lat: 35.379N

Lon: 78.734W

Elev: 198

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03712

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	26.5	-1.8	5.0	34.6	4.6	7.0	35.8	18.5	57.0	16.4	53.8	2.8	320	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	97.0	75.8	94.1	75.4	91.2	74.5	80.3	89.3	79.0	87.4	77.9	85.8	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	149.5	83.2	76.9	141.3	82.0	75.4	134.1	80.6	43.9	89.1	42.6	87.3	41.2	85.4	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	12.7	11.1	16.7	100.1	3.9	3.3	13.8	102.4	11.5	104.4	9.3	106.3	6.5	108.7
			14.3	82.7	3.8	2.5	11.6	84.5	9.4	86.0	7.3	87.4	4.5	89.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	43.8	46.7	53.7	63.5	71.3	78.3	81.1	80.2	73.6	63.1	52.5	47.1
	DBStd	15.34	10.10	9.30	9.47	8.21	6.68	5.24	3.57	4.38	5.62	8.16	8.39	9.20
	HDD50	727	247	160	72	6	0	0	0	0	0	6	67	169
	HDD65	2773	658	516	366	123	24	0	0	0	7	135	384	560
	CDD50	5475	56	67	187	412	662	850	965	935	709	411	142	79
	CDD65	2045	2	4	17	79	220	400	500	470	265	75	9	4
	CDH74	19103	12	27	165	793	2001	3949	4989	4523	2027	539	65	13
	CDH80	7890	1	2	21	226	728	1769	2306	2012	704	114	7	0
Wind	WSAvg	3.9	4.2	4.5	5.1	5.0	4.1	3.6	3.3	2.9	3.6	3.2	3.3	3.6
Precipitation	PrecAvg	47.2	3.7	2.9	3.9	3.2	3.8	4.5	5.8	5.2	4.7	3.3	3.1	3.0
	PrecMax	61.4	7.9	6.3	7.1	7.7	6.4	13.5	9.6	10.8	17.7	7.7	8.3	6.9
	PrecMin	33.5	1.1	1.0	0.9	1.0	1.1	1.4	1.7	1.4	0.4	0.0	0.4	0.9
	PrecStd	6.7	1.8	1.4	1.6	1.7	1.6	3.1	2.3	2.0	3.8	1.9	1.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.9	78.1	82.2	90.2	95.0	99.8	99.1	100.2	95.0	87.5	81.1	75.8
		MCWB	64.9	64.4	63.7	69.5	72.4	75.9	75.9	75.8	71.6	73.1	66.6	67.4
	2%	DB	69.6	72.6	78.9	85.8	90.3	95.6	96.6	95.4	90.2	82.8	75.1	71.5
		MCWB	62.0	60.3	62.2	66.3	70.9	75.2	76.7	75.7	73.4	69.4	65.5	64.6
	5%	DB	64.2	67.7	74.8	82.0	87.6	92.7	93.1	92.5	87.6	79.6	71.7	66.5
		MCWB	58.0	58.9	60.3	65.2	70.3	73.7	76.7	76.2	71.7	67.6	62.1	62.0
	10%	DB	60.9	63.6	70.3	79.0	84.0	89.6	90.7	89.9	83.9	76.8	66.6	63.2
		MCWB	53.9	54.2	59.1	63.6	68.8	73.1	75.3	75.3	71.4	67.2	60.4	57.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.7	67.8	70.0	75.5	79.1	80.7	82.0	82.2	78.9	76.8	71.8	70.0
		MCDB	71.5	71.9	76.6	83.0	85.4	91.2	92.3	90.9	86.2	79.5	76.5	71.9
	2%	WB	63.8	64.0	66.5	72.3	76.1	78.6	80.3	80.2	77.1	74.6	67.9	66.5
		MCDB	66.6	69.5	72.1	79.2	82.9	88.2	90.0	88.2	84.0	79.1	71.1	69.5
	5%	WB	60.3	60.6	64.2	69.6	73.7	77.2	78.9	78.7	75.5	72.2	64.7	63.2
		MCDB	64.1	66.2	69.6	76.6	80.9	85.4	87.8	86.4	81.1	76.8	68.8	65.7
	10%	WB	55.6	56.0	62.1	67.0	71.8	75.8	77.7	77.7	74.3	69.6	61.6	58.9
		MCDB	59.1	61.7	68.2	74.2	79.6	83.8	86.1	85.2	79.5	74.4	66.5	62.4
Mean Daily Temperature Range	5% DB	MDBR	19.0	20.5	22.2	23.0	20.9	19.3	18.5	17.9	18.1	20.6	21.6	19.3
		MCDBR	24.3	26.1	26.5	26.8	24.5	23.0	21.5	21.4	21.7	22.6	24.6	22.2
		MCWBR	17.3	16.4	13.2	11.1	8.4	7.3	7.0	6.7	7.6	11.1	15.0	16.0
	5% WB	MCDBR	21.0	23.1	21.1	21.2	20.3	19.1	19.0	17.9	16.1	16.1	19.4	19.5
		MCWBR	17.6	17.8	14.5	12.3	9.9	7.6	7.5	6.8	7.6	9.7	15.2	16.6
Clear-Sky Solar Irradiance	taub	0.322	0.338	0.371	0.413	0.467	0.509	0.555	0.534	0.432	0.388	0.341	0.322	
	taud	2.472	2.422	2.363	2.245	2.136	2.062	1.951	1.996	2.280	2.373	2.464	2.509	
	Ebn at Noon	279	286	285	277	262	250	238	240	262	265	269	272	
	Edh at Noon	27	32	37	43	49	53	59	55	40	33	27	25	
All-Sky Solar Radiation	RadAvg	831	1043	1379	1735	1914	2030	1929	1737	1462	1189	913	723	
	RadStd	55	120	98	118	145	137	117	89	119	154	73	60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+203

Nomenclature: See separate page